

The coal that has to go, by 2030

The question

You say you built an alignment methodology. Here it is on public inputs: how far does the fleet of listed European power utilities sit from an NGFS pathway?

The estimand

For a set of listed power utilities, the production-weighted gap between the capacity held in a technology in 2020 and the PACTA market-share target for it in 2030, as a share of the set's total capacity. Positive: more than the target. Capacity is the production metric, as in PACTA's power sector.

The answer

22 listed groups, 220,459 MW. Under NGFS Net Zero 2050 (REMIND, EU28) the set holds 38.6% of its capacity in coal against a 2030 target of 2.3%:

a coal gap of +36.2% [+24.4%, +48.3%] of the set's capacity.

Coal, gas and oil together: +59.4% [+47.5%, +69.9%].

The interval resamples the utilities, with the scenario held fixed.

Read this first

The gap is between the fleet as it stood in 2020 and the target. It is not a gap between the companies' plans and the target: the public asset register carries no retirement or build-out plans, so none enters. That makes it a measure of the change required, and it overstates misalignment wherever a utility has announced closures or a pipeline.

Method and inputs

Data, all fetched by `src/pacta_alignment.py` on each run

Plants and owners: WRI Global Power Plant Database, raw.githubusercontent.com.

9,519 plants in the EU27 and UK commissioned by 2020 or with no year, 700,450 MW.

Scenario: NGFS Phase 5, IIASA Scenario Explorer, anonymous API, Capacity|Electricity.

Three IAMs (REMIND, GCAM, MESSAGE), each on its own Europe region.

Listed status: Wikidata, a stock-exchange statement with no end date.

Validation: Eurostat `nrg_inf_epc`, EU27 net electrical capacity by fuel.

The target, from `r2dii.analysis (RMI)`

Increasing technologies (renewables, hydro, nuclear): the company adds the scenario's change as a share of the scenario's sector size, times its own sector capacity.

$$\text{target} = P0_tech + P0_sector \times (S_tech(t) - S_tech(0)) / S_sector(0), \text{ floored at } 0$$

Decreasing technologies (coal, gas, oil): it follows the scenario's ratio.

$$\text{target} = P0_tech \times S_tech(t) / S_tech(0)$$

The set

WRI names an owner for 456,552 of 700,450 MW. Owner strings map to 29 groups by a name rule; 22 are listed by Wikidata and above 2,000 MW.

Not listed by Wikidata: EDF, ESB, Edison, Statkraft, TVO, Vattenfall.

The set holds 220,459 MW, 31% of the register.

Allocation: ownership splits stated percentages; control gives a plant to its first-named owner.

Result: the scenario sets the size, the fleet sets the mix

What the chart shows

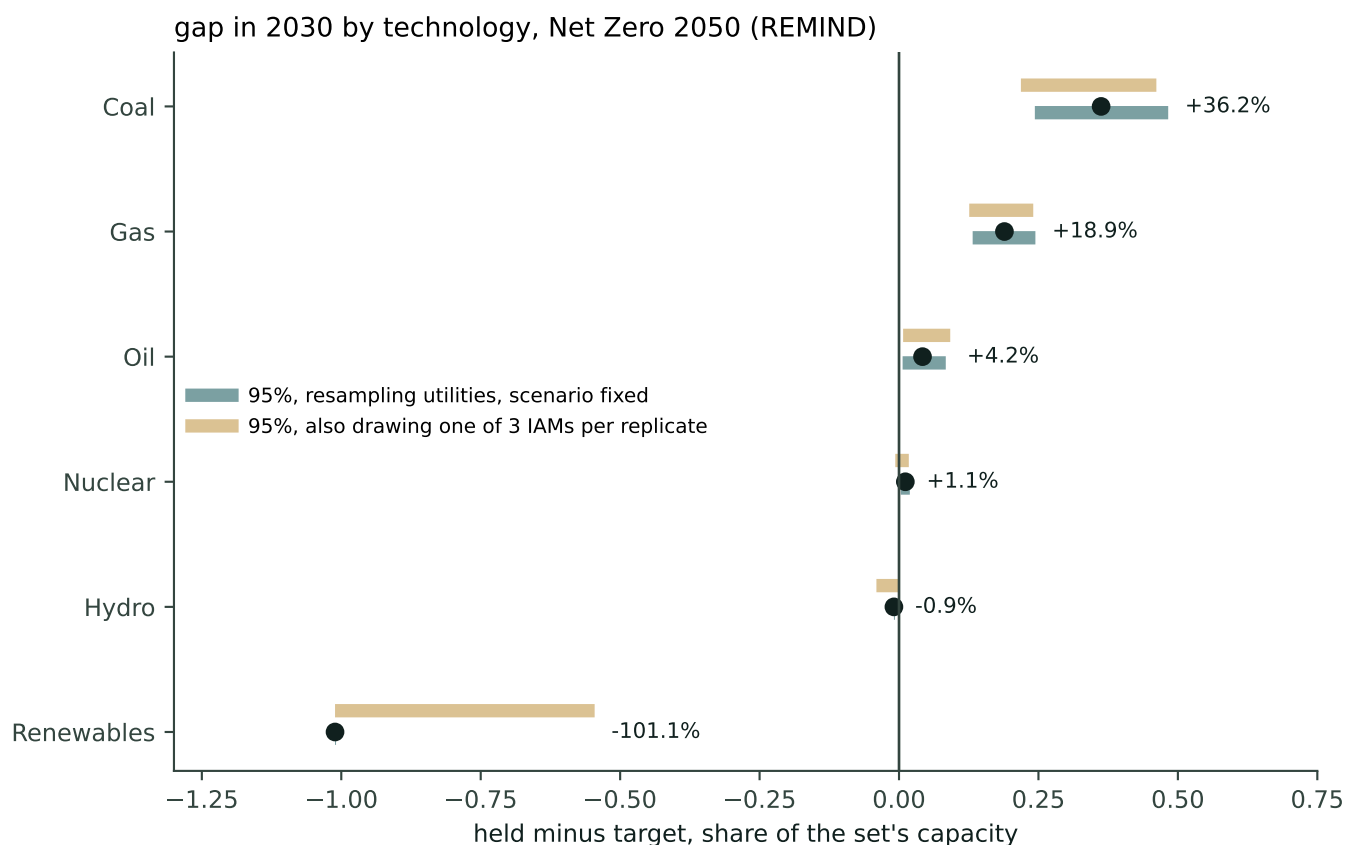
Coal +36.2%, gas +18.9%, oil +4.2%. Renewables -101.1%: the target is 108.5% of today's capacity against 7.4% held, because the scenario's EU renewable capacity changes by +269% to 2030.

Why the renewables interval has no width

Under the market-share approach every company is asked to add the same fraction of its sector. Summed over any set of companies, the build-out gap is minus the scenario's change over sector size, whoever is in the set. Company data cannot move it. Only the decreasing technologies depend on the companies: their gap is the held share times (1 - scenario ratio). The wider interval draws the scenario model once per replicate.

Who

The two largest groups hold 24.8% of the set's MW. Dropping any one group moves the coal gap between 33.1% and 40.6% (headline 36.2%).



Validation and backtest

Against a source this module did not fit

Eurostat's EU27 capacity by fuel checks the register. Coverage by technology, WRI / Eurostat:

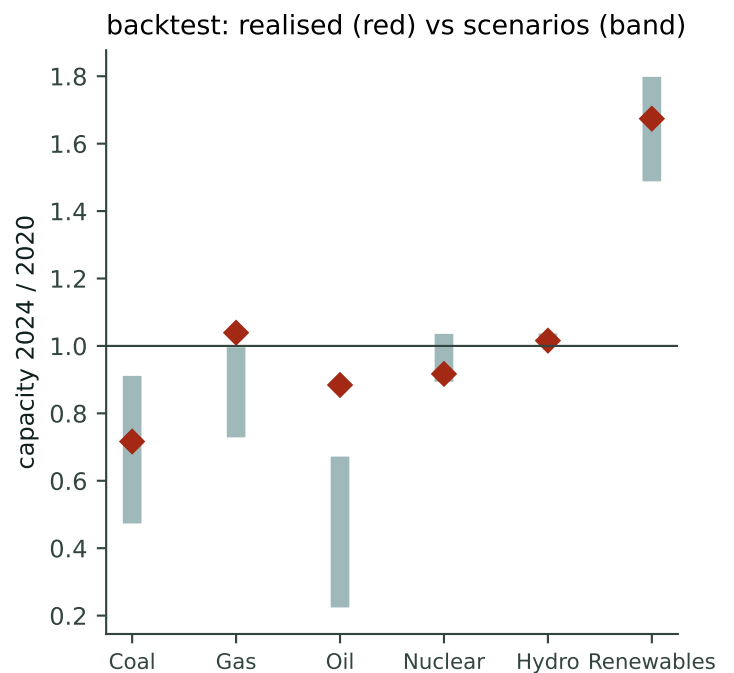
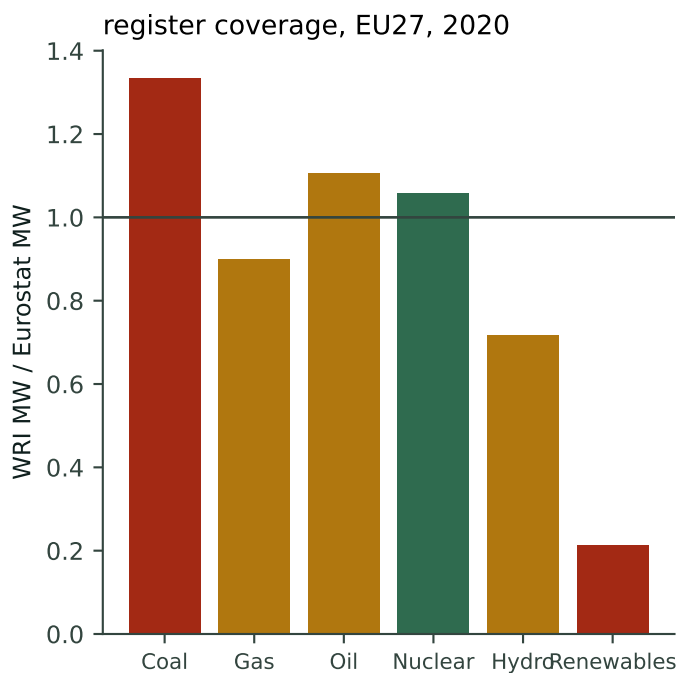
Coal 1.33, Gas 0.90, Oil 1.11, Nuclear 1.06, Hydro 0.72, Renewables 0.21.

Renewables are badly under-covered and coal over-covered. The register's latest commissioning year is 2018; I did not test whether closures by 2020 explain the coal excess. The set's mix is therefore tilted to fossil.

Rescaling each technology to Eurostat moves the coal gap from 36.2% to 21.6%.

Backtest of the scenario's first four years

Nothing here is forecast by me; the forward-looking object is the scenario. I test its 2020-24 path against what Eurostat reports for 2024. Coal capacity fell to 0.72 of its 2020 level; REMIND Net Zero had 0.47. Scenario errors by technology are in results/. Carrying the realised pace forward at a constant annual rate moves the coal gap to 32.0% and the renewables gap to -78.4%. No other source reports listed status, so that step has no independent check: it is Wikidata against itself.



Sensitivity: what moves the answer

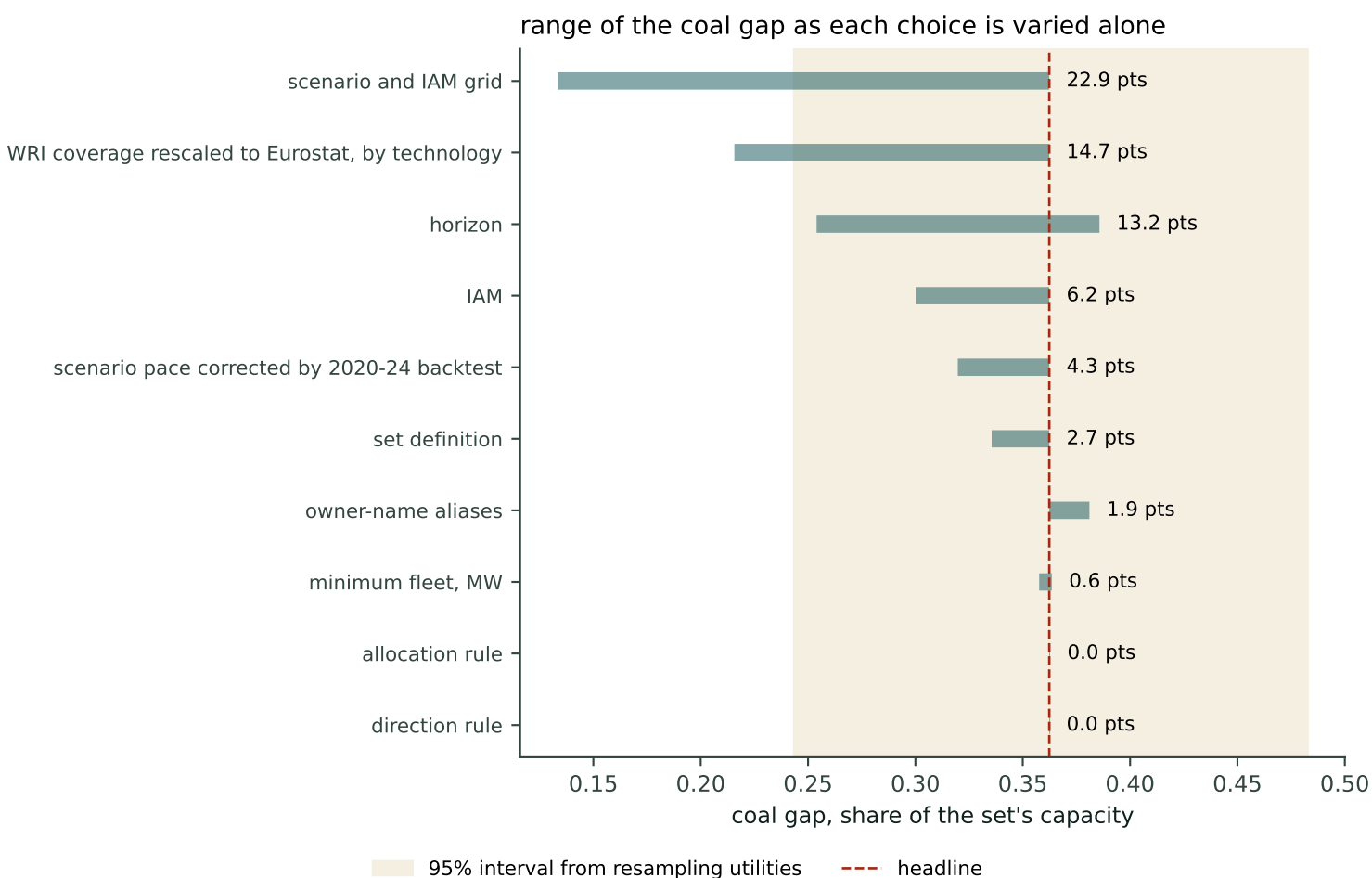
Ranked by the span of the coal gap

1. scenario and IAM grid: 22.9 pts.
2. WRI coverage rescaled to Eurostat, by technology: 14.7 pts.
3. horizon: 13.2 pts.

The headline sits at the top of the scenario grid (13.3% to 36.2% across 12 IAM-scenario pairs). The span of each choice is read off the coal gap alone.

The allocation rule, which practitioners argue about

It moves nothing here: 1.0% of attributed MW (3,006 MW) changes owner, and the rank correlation of company coal shares across the two rules is 0.999. WRI gives one owner string per plant and few joint ventures, so the rule has little to bite on. That is a property of the data, not evidence the rule is unimportant.



Limits, and what I would build next

The assumption that breaks the conclusion

The fleet is held at its 2020 level. If the utilities' announced retirements and build-out are what a plan-based PACTA run would use, the gaps here shrink, and the ranking of companies may change. WRI has no plans, so this module cannot size that. The coal gap is 36.2% of the set's capacity only if no coal capacity leaves before 2030.

What a sceptical reviewer attacks first

Coverage. The share-of-capacity form depends on the register's mix; rescaling to Eurostat gives 21.6%. The MW held in coal does not depend on the other technologies' coverage. Listing is as of today, owners as of the register's vintage. EDF's Wikidata record: Euronext Paris (until 2023-05-18). It is counted as unlisted.

Also

Three IAMs report different Europe regions; MESSAGE's is wider than the EU28 plant set. Eurostat hydro includes pumped storage; the NGFS hydro definition may not. Alias rules are my judgement; the strict tier is the sweep that bounds it.

Next

A plan-based run needs company retirement and pipeline data. I found no source of it that I could fetch by code without a login; I did not exhaust the options. That is the single input that would turn this fleet gap into the plan gap PACTA measures.